

Homework 1 — Solutions

Your Name

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Collaborators. Name everyone you discussed these problems with. Write “None” if you worked alone.

AI use declaration. Name the **tool**. Say **what it did** (drafted prose? checked algebra? wrote code? found references?). Say **what you verified, and how**. Say **what you did not verify**.

Claiming more checking than you actually did is worse than admitting the gap. An undeclared use, or an inaccuracy traced to one, scores zero.

If you used no AI tools, write: “No AI tools were used.”

Problem 1. Implementation and measurement

Code in the .py file. Tables, figures and answers here.

(a) Implementations and test results.

test	t	expected	linear	binary
first position				
last position				
interior position				
absent				
$n = 0$				
$n = 1$				
duplicates				

(b) Comparison counts at $n = 15$, and the two worst cases.

target t	0	1	2	3	4	5	6	7
linear								
binary								

Worst case, linear:

Worst case, binary:

n	1	2	3	4	7	8
worst case, binary						

(c) The n at which recursive linear first fails, the error, and the reason; whether recursive binary fails at $n = 2^{20}$; and the two loop versions.

```
def linear_loop(A, lo, hi, t):
    # your loop here

def binary_loop(A, lo, hi, t):
    # your loop here
```

(d) Timings, repetitions used and how you chose them, and the figure.

n	10^3	10^4	10^5	10^6	10^7
linear_loop					
binary_loop					

(e) Fitted slopes, expected slopes, what $\Theta(\log n)$ looks like on these axes, and any anomaly with its probable cause.

Slope, linear: Slope, binary:

Problem 2. Completeness: does the procedure halt?

(a) A measure for linear and one for binary: the quantity, why it is a non-negative integer, and why it strictly decreases.

Measure for linear: *Measure for binary:*

(b) The procedure down:

- (i) the $k \geq 0$ on which it halts, and the arguments for one k on which it does not;
- (ii) the requirement in the definition of a measure that fails, and the call at which it first fails;
- (iii) the one-line change, and a measure for your version.

(c) The modified binary does not halt: array, target, sequence of calls, and where the measure stops decreasing.

call	mid	measure
1		
2		
3		

(d) Boundary cases: length 0 and length 1, for both procedures.

(e) Two procedures:

- (i) halts on every input, not correct;
- (ii) correct whenever it returns, does not always halt.

Problem 3. Correctness: does it return the right answer?

- (a) The claim for `linear`, as one precise sentence covering both outcomes.
- (b) Proof for `linear`, by induction on d .

Base case.

Inductive hypothesis.

Inductive step.

- (c) The claim for `binary`, and its proof by induction on d . The step has three branches; handle all three.

Base case.

Inductive hypothesis.

Inductive step.

▷ **Where I got stuck:** If one branch defeated you, state which and how far you got.

- (d) The step that fails without sortedness, and an unsorted array of length 5 on which binary returns NOT-FOUND although t is present.

call	lo, hi	mid	branch taken
1			
2			
3			

- (e) Duplicates:

- (i) what each procedure returns, and whether the claim is violated;
- (ii) binary modified to return the smallest such index;
- (iii) the claim it satisfies, and which branch of the proof changes.

Problem 4. Time and space: what does the procedure cost?

- (a) Recurrences with base cases, why the floor, the solutions, the check against Problem 1(b), the values at $n = 10,000$, and the best cases.

$$T_{\text{lin}}(n) = \quad T_{\text{bin}}(n) =$$

n	1	2	3	4	7	8	15
counted in Problem 1(b)							
your formula							

- (b) $\mathbb{E}[C] = (n+1)/2$ for linear, in Θ terms, and whether averaging changes the choice.
- (c) Depths and extra space; the account of the Problem 1(c) failure; the space of the loop versions; and the comparison table.

	worst	best	space, recursive	space, loop	needs sorted
linear					
binary					

- (d) Sorted linked list: running time of each in Θ terms, the operation whose cost changed, and which procedure changes more.
- (e) The registry:

- (i) the two totals, q^* , and its values;
- (ii) cost of one insertion, locating separated from making room;
- (iii) the property a storage scheme would need for $O(\log n)$ throughout.

n	10^3	10^6	10^9
q^*			